

Auroral Records and Solar Cycle

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0.1 1. Point-process Periodogram

$$I(f) = \frac{1}{N} \left| \sum_{n=1}^N e^{2\pi i f t_n} \right|^2 = \frac{1}{N} \left[\left(\sum \cos 2\pi f t_n \right)^2 + \left(\sum \sin 2\pi f t_n \right)^2 \right] \quad (1)$$

This is the Bartlett (1963) spectral estimate for point processes, later re-derived by Fogel and Gavish (1988) for sparse timing data. It determines the frequency f at which the event times t_n align coherently in phase.

Key properties:

- No assumption about the form of $\lambda(t)$
- The 1535 CE burst contributes additional phase constraints without distorting the estimate
- Asymptotically consistent as $N \rightarrow \infty$
- Multimodal by construction due to finite observation window

With $T \approx 870$ years and $P \approx 10$ years, there are approximately 87 cycles, giving a theoretical resolution

$$\Delta P \approx \frac{P^2}{T} \approx 0.11 \text{ years.}$$

Result: Dominant peak at $P = 9.937$ years.

0.2 2. Modified Least-Squares (Quinn, Clarkson & McKilliam, 2012)

Each auroral event is treated as a noisy observation of an underlying periodic phase:

$$t_n = \alpha + \beta S_n + \varepsilon_n,$$

where $\beta = P$ is the unknown period, S_n are unknown integers representing cycle counts, and ε_n is timing noise.

The estimator minimises:

$$\text{SSMOD}(\gamma) = \min_{\nu} \sum_{n=1}^N \langle \gamma t_n - \nu \rangle^2 \quad (2)$$

where $\gamma = 1/P$ and $\langle x \rangle = x - \text{round}(x)$.

At the correct γ , the values $\{\gamma t_n\}$ cluster near a common phase ν , while incorrect values produce uniform scatter.

Key properties:

- Strongly consistent estimator ($\hat{\beta}_N \rightarrow \beta_0$)
- Convergence rate $N^{3/2}$
- No assumptions on noise distribution or observation gaps

Result: Minimum at $P = 9.919$ years.

The agreement between this method and the periodogram provides strong evidence for a genuine signal.

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0.3 3. Peak Refinement

The initial frequency grid has spacing

$$\Delta f = \frac{f_{\max} - f_{\min}}{n_f},$$

which corresponds to

$$\Delta P \approx P^2 \Delta f \approx 0.03 \text{ years.}$$

To improve accuracy, Brent's method is applied to $-I(f)$ within a window of $\pm 2/T$ around the dominant peak.

Result: Refined estimate $P = 9.937$ years.

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